

Supplementary

I. PRIVACY ANALYSIS

We measure the privacy of DSDPFed through two aspects of analysis: sensitive information and threat model.

Privacy analysis for Sensitive Information: To enhance the analysis of private information confidentiality in this paper, we conduct a detailed examination of the sensitive attributes in Table I. Precisely, we classify the sensitive attributes into three categories: 1) Client Identification (ID), 2) Upward Local Parameter (ω_{up}), and 3) Downward Global Parameter (ω_{down}). The trustworthiness of sensitive information primarily depends on whether it is secure for non-relevant observers (O). Comparing our DSDPFed with the curator model, local model, and FLAME [1] (the latest shuffler model scheme). FLAME's scheme satisfies (ϵ, δ) -DP for uploaded messages, making it trustworthy for any O . However, the client's identity is not trusted since the shuffler knows the source of the uploaded gradient. In our scheme, the decentralized Mixnets effectively restrict the collection of identity information by each mix node. FLAME fails to protect against global gradients, allowing the shuffler and attacker to obtain gradient information. This challenge is also present in our scheme. Thus, there is room for improvement in this area. Our next step involves considering the use of optimized encryption methods [2, 3] or exploring the possibility of the server broadcasting only a portion of the updated gradient. Overall, we have significantly improved the security of clients' identities and achieved high-level privacy.

Privacy Analysis for Threat Model: We analyze the privacy of the discussed models by evaluating the adversary's prior knowledge, potential attacks, and the limitations imposed by DSDPFed on adversary capabilities. Firstly, for the server, we assume it has access to supportive information and local updates. The server may attempt an inference attack against a single client. However, it cannot ascertain the client's identity even if it bypasses the differential privacy mechanism, thereby ensuring security for all clients. Next, for the shuffler, we employ multiple mix nodes that receive encrypted packets. Each node decrypts the packet using its private key and only obtains the location of the next hop, without accessing any raw client information. This method significantly limits the shuffler's ability to infer private data. Finally, global adversaries might infer private information through eavesdropping or data theft during transmission. Despite the protection mechanisms of encryption and differential privacy, consistent transmission patterns in updates could be vulnerable to traffic analysis. However, executing such an attack would require controlling the entire system, which is practically challenging.

It is essential to note that our analysis primarily considers

each adversary type independently (shuffler, server, and global adversary). If an adversary were to control both the shuffler and the server simultaneously, it could potentially reconstruct the identity of the clients by correlating the decrypted packet locations with the local updates. Addressing such combined threats requires more advanced mechanisms and is beyond the scope of this work.

II. PRIVACY AMPLIFICATION

In DSDPFed, we perform the algorithm R_l that satisfies (ϵ_l, δ_l) -LDP, thus providing a local privacy guarantee of ϵ_l for each gradient of the upload. Subsequently, we will analyze the global privacy guarantee (ϵ, δ) - DP satisfied by the scheme and prove that our scheme has a double privacy amplification effect.

We introduce a perspective *View* $\mathcal{V}$. And we define two viewpoints $\mathcal{V}_c, \mathcal{V}_g$ to denote the client-side view and the global view, respectively. Where ϵ_l, ϵ_c privacy guarantees for perturbations are implemented under the client perspective $\mathcal{V}_c$, i.e., the DP guarantees for privacy information is always satisfied for different phases of any client i : $\Pr_{\mathcal{V}_c}[\mathcal{R}(D) \in S] \leq e^{\epsilon_l, c} \Pr_{\mathcal{V}_c}[\mathcal{R}(D') \in S]$. Similarly, ϵ is a privacy guarantee from the global perspective, satisfying $\Pr_{\mathcal{V}_g}[\mathcal{R}(D) \in S] \leq e^\epsilon \Pr_{\mathcal{V}_g}[\mathcal{R}(D') \in S]$. In DSDPFed, privacy amplification is specified in the mapping of $\epsilon_l \rightarrow \epsilon_c \rightarrow \epsilon$, which means that we can achieve stronger privacy guarantees globally $\epsilon(\epsilon < \epsilon_l)$ with the same privacy guarantee ϵ_l .

Lemma 1: (Privacy amplification proved by shuffling [4]): Assume that the local private algorithm R satisfies (ϵ_l, δ_l) -LDP with $\epsilon_l \leq \frac{\log(qn/\log(1/\delta))}{2}$ and $\delta_l > 0$. Also $q = \frac{ks}{Nr}, B > 0$, for any T and i , M_i^T satisfying $(\tilde{\epsilon}, \tilde{\delta})$ -DP.

$$\tilde{\epsilon} = \mathcal{O} \left(\min \{ \epsilon_l, 1 \} e^{\epsilon_l} \sqrt{\frac{\log(1/\tilde{\delta})}{qn}} \right) \quad (1)$$

Here, M_T is noted as the global private algorithm, including local perturbation and shuffle. $M_T(D)$ can be defined as: $M_T(D) = \mathcal{S} \circ \mathcal{R}(D_i)$. The analysis of M_T can represent the privacy guarantee from the global perspective.

Local privacy amplification is achieved by two aspects: sample and shuffle. Equation 1 denotes privacy amplification by shuffling [5]. That is, for the shuffle-only model, if $\epsilon_l \leq \frac{\log(qn/\log(1/\delta))}{2}$, $\delta_l > 0$, the algorithm M_T satisfies $(\tilde{\epsilon}, \tilde{\delta})$ - DP. Furthermore, if $\epsilon_l = \mathcal{O}(1)$, then $\tilde{\epsilon} = \mathcal{O}(\epsilon_l \sqrt{\frac{\log(1/\tilde{\delta})}{qn}})$.

TABLE I: Privacy analysis for sensitive information (ω_{up} , ω_{down} and ID) across different models and adversary types

Parties	Shuffler			Server			Global Adversary		
Sensitive Info	ω_{up}	ω_{down}	ID	ω_{up}	ω_{down}	ID	ω_{up}	ω_{down}	ID
The Curator Model	N/A	N/A	N/A	✗	✓	✗	✗	✗	✗
The Local Model	N/A	N/A	N/A	✓	✓	✗	✓	✗	✗
FLAME	✓	✗	✗	✓	✓	✓	✓	✗	✗
DSDPFed	✓	✗	✓	✓	✓	✓	✓	✗	✓

Lemma 2: (Privacy amplification proved by sampling [6]) Assuming $R(D)$ satisfies $(\epsilon, \delta) - DP$, with subsampling $D' = \text{sample}(D, q)$, for any $M'(D')$ satisfies:

$$\epsilon' = \ln(1 + q(e^\epsilon - 1)) \quad (2)$$

In particular, when $\epsilon < 1$, M' satisfies $(O(q\epsilon, q\delta)) - DP$

Sampling likewise achieves privacy amplification, which can be refined into client sampling and local training samples, denoted by k and s , respectively. The sampling probability $q = \frac{ks}{Nm}$. Unlike the previous method that randomly samples k users, in DSDPFed, we can sample the aggregation of users that are more valuable for model convergence by value-based threshold screening, and the sampling k' can be optimized as $k' = \text{sample}(\gamma, c)$. The sampling probability can be optimized as $q' = \frac{k's}{N_r}$. In Lemma 2, if $s = 1$ and $\epsilon_l = \mathcal{O}(1)$, we have $\epsilon_c = \ln(1 + q(e^\epsilon - 1))$. Nevertheless, the formula constraints to uploading only one compressed gradient per client per round. However, for training in complex scenarios ($s > 1$), Girgis [4] et al. correct the privacy amplification to $\epsilon_c = \ln(1 + \bar{q}(e^{\bar{\epsilon}} - 1)) = \mathcal{O}(\bar{q}\bar{\epsilon})$, and $\delta_c = \bar{q}\bar{\delta}$, where $\bar{q} = q_2/q_1$. In DSDPFed, each client only uploads one model gradient information ($s = 1$), so our proof follows Equation 2. By substituting the $\bar{\epsilon}$ in Equation 1 and using $ks = qn$, if $\epsilon_l = \mathcal{O}(1)$, we can get an double amplified ϵ_c privacy.

$$\epsilon_c = \mathcal{O}(\epsilon_l \sqrt{\frac{q' \log(1/\bar{\delta})}{n}}) \quad (3)$$

Lemma 3 (Strong composition [7]): Let $M_1, \dots, M_t$ denote t local perturbation algorithms satisfying $(\epsilon_c, \delta_c) - DP$, where $\epsilon_c > 0$, $\delta_c > 0$. Then for their combined algorithm $M = (M_1, \dots, M_t)$, for any $\delta' > 0$ satisfying $(\epsilon, \delta) - DP$, where

$$\epsilon = \sqrt{2t \log(1/\delta')} \epsilon_c + t \epsilon_c (e^{\epsilon_c} - 1), \quad \delta = t \delta_c + \delta' \quad (4)$$

In particular, when $\epsilon_c = \mathcal{O}(\sqrt{\frac{\log(1/\delta')}{t}})$, we can calculate $\epsilon = \mathcal{O}(\epsilon_c \sqrt{t \log(1/\delta')})$

For t local perturbations satisfying ϵ_c , we can combine the analysis to obtain the global privacy guarantee ϵ . Lemma 2 supports a strong composition theorem. When performing aggregation, the privacy parameter of the composition mechanism scales as $\sqrt{t}$ with some loss in δ . However, the basic composition theorem in which the privacy parameter of the

composition mechanism scales as t . By substituting ϵ_c in Equation 3, we can calculate the global privacy parameter:

$$\epsilon = \mathcal{O}\left(\epsilon_l \sqrt{\frac{q' t \log(1/\bar{\delta}) \log(1/\delta')}{n}}\right) \quad (5)$$

By setting $\bar{\delta} = \frac{\delta}{2q't}$ and $\delta' = \frac{\delta}{2}$, we get

$$\epsilon = \mathcal{O}\left(\epsilon_l \sqrt{\frac{q' t \log(2q't/\delta) \log(2/\delta)}{n}}\right) \quad (6)$$

Proposition 1: (Privacy amplification by DSDPFed): Assuming sampling rate $q' = \text{sample}(\gamma, c)$, which is bounded by random rate and subsample rate in DSDPFed. So we amend the amplification bound as:

$$\epsilon = \mathcal{O}\left(\Gamma \epsilon_l \sqrt{\frac{c \Gamma s t \log(2c \Gamma s t / m \delta) \log(2/\delta)}{mn}}\right) \quad (7)$$

Here the rate of a sampled point is determined by response rate γ and threshold scale c . Here the γ is not a constant, but a distribution Γ bounded by the update norm. Therefore, the local random budget needs to be amended to $\epsilon_l \Gamma$. Here, we select the c -th highest norm as the sampling threshold and amend the sample rate as $\frac{c \Gamma s}{m}$.

III. ABLATION STUDY

A. Impact of ϵ

Different privacy budgets have a profound impact on model convergence, as evidenced by the recorded test results in Table II for cases where $\epsilon > 0$. Within the DSDPFed scheme, the MNIST-2NN dataset with independent and identically distributed (IID) data partitioning achieves a remarkable convergence accuracy of 96.727% when ϵ is set to 6. Conversely, the accuracy dramatically declines to 83.255% when subjected to ultra-privacy conditions with $\epsilon = 0.1$. In scenarios involving non-independent homogeneous distributions or complex CIFAR10 datasets, we observe that excessively tight privacy budgets hindered convergence, resulting in unsatisfactory accuracy or NaN values exceeding permissible bounds. Nevertheless, DSDPFed consistently demonstrates remarkable utility across various security scenarios. We observe that reducing the privacy budget correspondingly reduces the accuracy under all conditions. This can be attributed to the amplified deviation of post-perturbation outcomes from actual values when ϵ assumes a smaller value, yielding more security updates. However, the aggregation model's accuracy is relatively compromised due to

the cumulative effect of heightened perturbation. This observation reinforces the inherent trade-off between privacy and utility, wherein our DSDPFed scheme effectively addresses this trade-off, leading to utility enhancements compared to traditional approaches.

ϵ	0.1	0.5	1.0	6.0
MNIST-2NN, IID	83.524%	96.028%	96.393%	96.727%
MNIST-CNN, NIID	10.3%	90.658%	94.528%	96.727%
CIFAR-CNN, IID	NaN	56.784%	75.614%	79.835%

TABLE II: The performance of DSDPFed tested in different scenarios with different privacy budgets ϵ .

B. Impact of γ

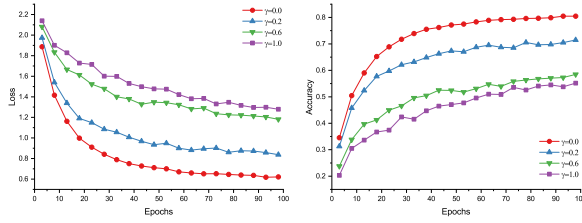


Fig. 1: In the condition of CIFAR-CNN, IID, $\epsilon = 0.5$, test the utility of $\gamma = 0.0, 0.2, 0.6, 1.0$ respectively. The top graph shows the test accuracy on the test set, and the bottom graph shows the variation of loss values in training.

The choice of γ value plays a crucial role in the dynamic DP design of DSDPFed. However, it is essential to investigate how different preset values impact experimental outcomes. When $\gamma = 0$, indicating no perturbation at all (no-DP), and $\gamma = 1$, representing full perturbation (full-DP), it is evident from Fig. 1 that on the CIFAR10 dataset, smaller perturbation probabilities correspond to higher test accuracy and smaller loss values. This highlights the significant influence of the chosen γ value on experimental results. With the same differential privacy budget, smaller differential probabilities result in fewer users contributing perturbation, leading to relatively higher model accuracy but reduced security. In other words, smaller differential probabilities are suitable for scenarios with more relaxed security requirements and stringent accuracy requirements. Conversely, larger differential probabilities suit scenarios prioritizing high security and accepting lower accuracy. Additionally, we observe that various factors, including local update sizes and different datasets, influence our dynamic γ value. In our experiments, we ensure that $0.5 < \gamma < 0.8$, which ensures that a significant portion of the data is protected while balancing privacy and utility.

C. Impact of ξ and c

In client subsampling, we find that the choice of sampling probability ξ and threshold scale c affects the value-based

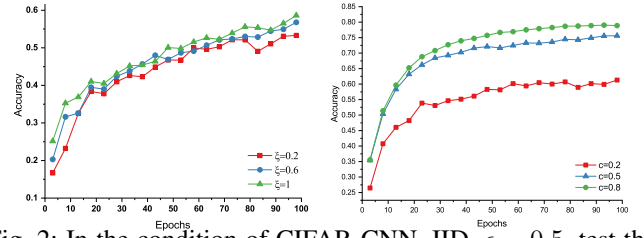


Fig. 2: In the condition of CIFAR-CNN, IID, $\epsilon = 0.5$, test the accuracy at the preset client sample rate $\xi = 0.2, 0.5, 0.8$, Threshold Scale $c = 0.2, 0.5, 0.8$, respectively. A value of 1.0 means that all clients participate in filtering, but not all participate in aggregation.

threshold selection and, thus, the number of aggregated users per round. In Fig. 2a, we fix c constant and preset different sampling probabilities ξ in CIFAR-CNN. When sampling fewer clients ($\xi = 0.2$), the model accuracy converges to 53.294%. And when all users participate in the ranking ($\xi = 1$), the accuracy can be improved to 58.834%. Under $\epsilon = 0.5$ strong perturbation, the influence interval is controlled around 5%. Fig. 2b presents the impact of c taking values. We observe that a larger c leads to higher accuracy, and the convergence results tend to be smoother. At the same time, too small c leads to a decrease in accuracy and a large fluctuation in the results. Since the number of users involved in aggregation in DSDPFed is determined by the threshold thr , the result shows that the threshold value in each round is affected by both sampling and scale. The more samples are sampled, the larger the threshold scale selected, the more clients thr , allowing more clients to participate in training, and the higher the training progress of the model. However, this also brings a higher communication cost. Furthermore, we find that the choice of c has a more significant impact on the experimental results than ξ , which means we can sample fewer samples and choose a larger scale to obtain a cut threshold that satisfies the communication cost and utility contradiction.

D. Impact of Clipping norm

In traditional DP-FL, it's customary to apply gradient clipping (C) prior to perturbation. This practice aims to mitigate the magnitude of noise required, yet inevitably influences the outcomes of our approach. In Figure.3, we explore this aspect of the impact. From the first subfigure.3a, the total norm of updates in a single round exhibits a direct correlation with the clipping value, thus affirming its influence on the overall perturbation. This association stems from the positive correlation between norms and the sensitivity of DP. From the next subfigure.3b, we find that too large or too small C is disadvantageous. The former introduces excessive noise, while the latter compromises the representational capacity of the target model. Consequently, the adaptive balancing of the clipping threshold warrants further investigation.

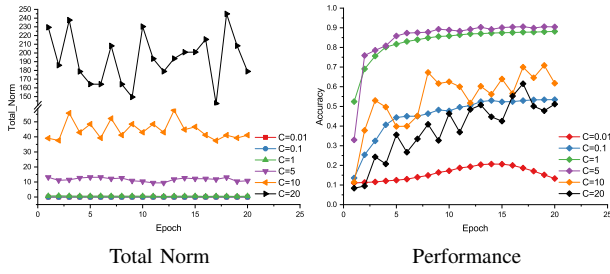


Fig. 3: In the condition of MNIST-2NN, IID, compare the total norm and accuracy performance of the per update round in different C .

REFERENCES

- [1] R. Liu, Y. Cao, H. Chen, R. Guo, and M. Yoshikawa, “Flame: Differentially private federated learning in the shuffle model,” in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 35, no. 10, 2021, pp. 8688–8696.
- [2] C. Zhang, S. Li, J. Xia, W. Wang, F. Yan, and Y. Liu, “{BatchCrypt}: Efficient homomorphic encryption for {Cross-Silo} federated learning,” in *2020 USENIX Annual Technical Conference (USENIX ATC 20)*, 2020, pp. 493–506.
- [3] Z. Jiang, W. Wang, and Y. Liu, “Flashe: Additively symmetric homomorphic encryption for cross-silo federated learning,” *arXiv preprint arXiv:2109.00675*, 2021.
- [4] A. Girgis, D. Data, S. Diggavi, P. Kairouz, and A. T. Suresh, “Shuffled model of differential privacy in federated learning,” in *International Conference on Artificial Intelligence and Statistics*. PMLR, 2021, pp. 2521–2529.
- [5] B. Balle, J. Bell, A. Gascón, and K. Nissim, “The privacy blanket of the shuffle model,” in *Advances in Cryptology—CRYPTO 2019: 39th Annual International Cryptology Conference, Santa Barbara, CA, USA, August 18–22, 2019, Proceedings, Part II 39*. Springer, 2019, pp. 638–667.
- [6] S. P. Kasiviswanathan, H. K. Lee, K. Nissim, S. Raskhodnikova, and A. Smith, “What can we learn privately?” *SIAM Journal on Computing*, vol. 40, no. 3, pp. 793–826, 2011.
- [7] C. Dwork, A. Roth *et al.*, “The algorithmic foundations of differential privacy,” *Foundations and Trends® in Theoretical Computer Science*, vol. 9, no. 3–4, pp. 211–407, 2014.